Aman Verma

SENIOR BACKEND ENGINEER

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TECHNICAL SKILLS

Languages & Core: Python, Go (Golang), Rust, TypeScript, C++, SQL (PostgreSQL), Bash

Cloud & Infrastructure: Docker (Containerization), Kubernetes (K8s), AWS, Terraform (IaC), Git

Databases & Storage: PostgreSQL, Redis (In-Memory Cache), SQLite (FTS5), ChromaDB

Frameworks & Developer Tools: FastAPI, Pydantic AI, PyTorch, Celery, Hugging Face, pytest

WORK EXPERIENCE

Software Engineering Intern – **LexoLegal** June 2024 -- Aug. 2024
Remote

- Architected multi-tenant legal search engine in **Python** using **SQLite FTS5** and **FastAPI**, reducing query latency from **850ms to 110ms** across **40,000 regulatory documents**.
- Engineered asynchronous worker queue with **Redis** distributed caching and **Celery**, processing **15,000 document transformations daily** with **sub-200ms latency**.

PROJECTS & OPEN SOURCE

ncert-rag-evals | *Python, PyTorch, ChromaDB, Hugging Face, Docker, pytest* | https://github.com/amanverma-765/ncert-rag-evals Sept. 2024 -- Present

- Architected automated evaluation framework across **84 unit and integration test suites** using **pytest**, achieving **350 evals/sec** throughput (**p < 0.01**).
- Engineered semantic chunking pipeline in **ChromaDB** with **Hugging Face** embeddings, reducing vector search latency by **42%** across **12,000 document chunks**.

doubtless | *Python, Pydantic AI, PostgreSQL, SQLite, Docker* | https://github.com/amanverma-765/doubtless Jan. 2024 -- May 2024

- Developed local-first hybrid retrieval engine fusing BM25 with dense embeddings via Reciprocal Rank Fusion, achieving **86.9% Recall@5 in 8.9ms on CPU**.
- Secured multi-tenant runtime isolation by binding tenant context inside **Pydantic AI** dependencies, preventing prompt-injection state leakage across sessions.

EDUCATION

Indian Institute of Information Technology 2021 -- 2025
Bachelor of Technology in Computer Science and Engineering *India*

Cumulative GPA: 8.6 / 10.0